A Case Study of Evasion Attack Against Audio Watermarking Models

Maharani Irawan
University of Chicago

William Nixon
University of Chicago

Yining Shi
University of Chicago

***Abstract*—Audio watermarking is increasingly used to detect synthetic speech and reduce misuse of generative voice models. However, the robustness of current watermarking systems under adversarial manipulation remains unclear. In this work, we evaluate the robustness of two watermark detectors under different attack settings: AudioSeal in a white-box setting with access to the detector model, and the ElevenLabs V2 detector in a black-box setting accessible only through its API. We study three types of attacks: simple audio processing transformations (e.g., pitch, speed, and sample/bit rate changes), a gradient-based adversarial attack (PGD), and a autoencoder-based purification attack. Experiments are conducted on multiple speech datasets and synthetic voice samples, evaluating both detector confidence and perceptual audio quality using the PESQ metric.**

Our results show that watermark detectors can be weakened with small modifications that preserve audio quality. PGD is highly effective in the white-box setting but transfers poorly to the black-box detector. In contrast, the autoencoder attack generalizes better across models. Surprisingly, even simple audio edits pitch adjustments can already reduce detection performance, despite being inexpensive and widely available in common audio editing tools. These findings suggest that current audio watermarking systems remain fragile under realistic audio editing pipelines and highlight the need for more robust watermarking techniques.

I. Introduction

The rapid growth of generative machine learning has made it increasingly easy to create realistic synthetic speech at scale. While this progress enables many useful applications, it also raises serious concerns about misuse, authenticity, and copyright protection [3], [6]. Synthetic audio can be used to imitate public figures, spread false information during sensitive events, or reproduce voices and content without consent. As a result, audio watermarking has emerged as an important tool for identifying whether a piece of audio is machine-generated or protected by copyright [6].

However, the robustness of audio watermarking remains much less understood than that of image watermarking. Images and audio differ in important ways. Audio is routinely modified before it reaches the listener: it may be compressed, resampled, denoised, sped up, pitch-shifted, enhanced, mixed with background sounds, or re-recorded as part of normal editing and distribution pipelines. At the same time, audio watermarking methods often need to embed low-energy signals in order to remain imperceptible. This creates a fundamental tension: the watermark must be strong enough to survive realistic transformations, yet subtle enough not to harm audio quality. In practice, this raises a basic question: *how robust are current audio watermarking systems under realistic perturbations?*

Motivated by this gap, we study the robustness of modern audio watermarking systems under both white-box and black-box attack settings. We consider *AudioSeal* in a white-box setting, where the detector model is accessible, and the *ElevenLabs V2* detector in a black-box setting, where only API access is available. We evaluate multiple attack strategies, ranging from simple and inexpensive audio manipulations to optimization-based and learned attacks.

Our results show that current audio watermarking systems are vulnerable even to relatively small modifications that preserve intelligibility and reasonable perceptual quality. White-box attacks can strongly reduce detector confidence on the target model, while learned black-box attacks transfer better across systems. More surprisingly, we find that simple audio edits such as small speed or pitch changes can already degrade detection performance. These attacks are cheap, widely available in standard editing tools, and require little technical expertise, making them practical in real-world settings.

Overall, this paper argues that robustness claims for audio watermarking should be evaluated under more realistic post-processing conditions. Our findings highlight common failure modes in existing systems and suggest that building robust audio watermarks may be fundamentally more challenging than current evaluations imply. We hope this study helps motivate more realistic benchmarks and stronger watermarking methods for synthetic audio.

II. Background

Recent industry and research efforts have proposed machine learning based watermarking systems for audio. In particular, *AudioSeal* by Meta is presented as a state-of-the-art and locally deployable watermarking system, with claims of robustness against common audio edits such as compression, re-encoding, and noise addition while maintaining minimal impact on perceptual quality [6]. In parallel, commercial speech generation platforms such as *ElevenLabs* have introduced their own detection systems to identify audio generated by their models. These systems are becoming increasingly important as watermarking is positioned as a practical defense for attribution, copyright protection, and deepfake detection.

AudioSeal's local watermark generator operates as an encoder and a decoder, both derived from EnCodec's design—a neural audio compression model [1]. The generator was trained to take a waveform as input and produce a watermark of the same dimensionality, where the audio is watermarked by adding the watermark to the original audio. To find the optimal watermark for the given input with minimal loss of audio quality, the generator was trained through optimizing the *Perceptual Loss*, the watermark's imperceptibility to the human ears. In unison, the detector is trained on a localized loss, of

which the process ensures that the detection of watermarked audio is done at the level of an individual sample. It serves as a useful white-box for researchers to test the robustness of the methodology, and has proven useful for watermarking average-use audio with minimal loss [6].

In contrast, little public information is available about ElevenLabs' proprietary *Eleven* speech generation and detection models. For the purpose of this study, we do not assume knowledge of their exact internal design. Instead, we hypothesize that they may rely on signal characteristics that overlap with those used by systems such as AudioSeal, and we use this setting to evaluate whether attacks developed on one model can transfer to another.

III. Evaluation Design

We evaluate the robustness of the watermarking systems using three classes of adversarial attacks. First, we apply simple audio editing transformations to both the white-box and black-box models, including changes in bitrate, sample rate, playback speed, and pitch. Second, we implement a white-box Projected Gradient Descent (PGD) attack on AudioSeal, which directly perturbs the waveform using gradient information from the detector. Third, we train an autoencoder-based purification model that performs waveform-level manipulation and targets the black-box Eleven V2 speech generator and detector.

For the white-box setting, we use 100 audio clips from the ML Commons People's Speech dataset [2] and 100 audio clips from the RAVDESS Emotional Speech Audio dataset. For the black-box setting, we generate 60 synthetic speech clips using the Eleven V2 model from 20 publicly available AI voice actors. These clips are used to implement and evaluate attacks against the ElevenLabs detector.

After each attack, we evaluate both attack effectiveness and perceptual audio quality. Attack effectiveness is measured using detector confidence, while audio quality is measured using the Perceptual Evaluation of Speech Quality (PESQ) [4] metric, which ranges from -0.5 to 4.5, where 4.5 indicates very high perceptual quality and lower scores indicate heavily degraded, poor-quality speech. We release our source code publicly at *https://github.com/williamnixon20/adverserial_ml*.

A. Audio Editing Perturbation

For straightforward audio perturbation, we propose four main methods:

- ***Bit Rate*** Degrading the bit rate—the amount of audio data transferred per unit of time (e.g. 128k to 64k).
- ***Sample rate*** Degrading or upscaling the sample rate—the number of times per second a piece of analog audio is measured (e.g., 16kHz to 12kHz or 16kHz to 48kHz).
- ***Playback Speed*** Slowing audio speed by a certain percentage (e.g., 0.9x).
- ***Playback Speed with Pitch Scaled*** Scaling/normalizing the frequency of the slowed audio speed by a certain percentage (e.g., 0.9x).

We implement all of the aforementioned attacks on collected audio pieces—against both AudioSeal and Eleven V2—to verify their claims of robust protection of the watermarks against common audio editing evasions. In Section IV, we show that both models fail to comprehensively protect their respective watermarks, given an acceptable PESQ (i.e., audible to human ears with reasonable levels of noise) perturbation budget.

B. PGD Attack

To evaluate the robustness of AudioSeal under a white-box adversarial setting, We implement a Projected Gradient Descent (PGD) attack that directly perturbs the audio waveform in the time domain. This is attack assumes a white-box approach with access to AudioSeal's watermark detector model. The objective of this attack is to reduce the watermark detector confidence while maintaining perceptual audio quality.

Unlike signal-processing attacks such as bitrate compression or speed perturbation, PGD operates at the level of individual waveform samples and optimizes perturbations through gradient descent. To illustrate, let $x \in [-1, 1]^T$ denote the audio waveform containing T time samples. In digital audio, each sample represents the amplitude of the signal at a particular time index determined by the sampling rate. The adversarial attack therefore perturbs the *amplitude of every audio sample*, producing an adversarial waveform x_{adv}.

The perturbation is constrained within an L infinity ϵ-ball around the original signal:

$$\|x_{adv} - x\|_\infty \leq \epsilon \tag{1}$$

where ϵ controls the maximum amplitude change applied to any individual sample. This constraint ensures that the perturbations remain small and largely imperceptible.

Essentially, PGD exploits highly guided small waveform perturbations to reduce detector confidence while maintaining perceptually similar audio. Let $C(x)$ denote the watermark detector confidence that the input audio contains a watermark. The attack aims to minimize this confidence score:

$$\min_{x_{adv}} C(x_{adv}) \tag{2}$$

Starting from the original waveform x, PGD iteratively updates the adversarial signal using the gradient of the detector confidence with respect to the waveform:

$$x_{t+1} = \Pi_\epsilon \left(x_t - \alpha \cdot \text{sign} \left(\nabla_x C(x_t) \right) \right) \tag{3}$$

where α is the step size and Π_ϵ denotes projection (clipping) back into the ϵ-bounded perturbation region around the original signal. After each update step, the waveform is clipped to the valid audio amplitude range $[-1, 1]$.

To ensure that adversarial perturbations do not overly degrade perceptual quality, we impose a perceptual perturbation

budget measured using the Perceptual Evaluation of Speech Quality (PESQ) metric. PESQ measures perceptual similarity between the adversarial audio and the original signal. During optimization, the attack is terminated if the PESQ score falls below a predefined threshold, ensuring that the modified audio remains intelligible.

C. Autoencoder Attack

To remove AudioSeal watermarks while preserving perceptual audio quality, we design a convolutional autoencoder that learns a compressed latent representation of the waveform and reconstructs a purified signal. The objective is to suppress watermark artifacts while maintaining the semantic and perceptual content of the original audio. Formally, given an input waveform x, the encoder $E(\cdot)$ produces a latent representation:

$$z = E(x) \tag{4}$$

where z denotes the compressed latent representation. The decoder $D(\cdot)$ then reconstructs the purified waveform:

$$\hat{x} = D(z) \tag{5}$$

where $\hat{x}$ represents the reconstructed audio signal.

The model is trained using AudioSeal watermarked audio as input and the corresponding clean (unwatermarked) audio as the target. We do not include a detector loss because we assume a black-box setting where the detector model and its gradients are not accessible. As a result, the autoencoder cannot be directly optimized to minimize detector confidence during training. The training objective combines waveform, spectral, and perceptual losses so that the model learns to suppress watermark artifacts while maintaining audio quality. The details of the losses are as follows:

a) Waveform Reconstruction Loss: A waveform L1 loss is used to enforce time-domain reconstruction accuracy:

$$L_{\text{wave}} = \|x - \hat{x}\|_1 \tag{6}$$

This loss ensures the reconstructed signal remains close to the original waveform.

b) Multi-Resolution STFT Loss: To preserve spectral structure across multiple frequency scales, we use a multi-resolution STFT (MRSTFT) loss:

$$L_{\text{mrstft}} \tag{7}$$

This loss measures discrepancies between spectrogram magnitudes computed at multiple window sizes. In training, the MRSTFT loss is assigned a weight of 0.5. To choose this value, due to resource bottleneck, we did a small scale hyperparameter tuning by monitoring the audio PESQ quality after 1 epoch of training with different weight for this loss metric.

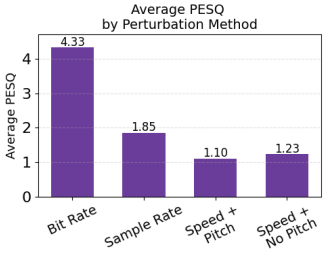

Fig. 1. Average PESQ score by perturbation method.

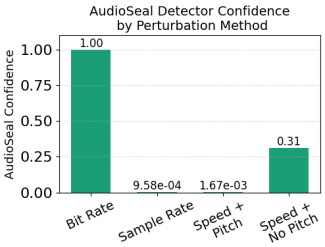

Fig. 2. Average AudioSeal detector confidence by perturbation method.

c) SI-SDR Loss: To further improve reconstruction fidelity, we incorporate the scale-invariant signal-to-distortion ratio (SI-SDR) loss:

$$L_{\text{SI-SDR}} \tag{8}$$

SI-SDR directly optimizes signal quality and separation performance. Empirically, this loss plays a major role in improving perceptual reconstruction quality. The SI-SDR loss is applied with a weight of 0.01. This value is chosen because the SI-SDR output ranges between 0-20, hence has to be scaled down to avoid overshadowing other loss.

The final training objective combines all loss terms:

$$L = L_{\text{wave}} + 0.5L_{\text{mrstft}} + 0.01L_{\text{SI-SDR}} \tag{9}$$

The combination and weights of these parameters are empirically chosen to minimize detector confidence while still preserving audio quality. We chose a combination of perceptual losses since we found that no single loss was sufficient to acceptably preserve audio quality. In the design of our autoencoder, we only use PESQ as a metric to measure audio quality. It is not directly incorporated in our loss function due to the non-differentiable nature.

IV. Result

A. Audio Editing Perturbation

As shown in Figure 2, while Bit Rate perturbation had almost no effect on the detector confidence for the AudioSeal watermark, all other three—particularly Sample Rate and Playback Speed perturbations—have effectively collapsed the AudioSeal watermark detector. Respectively, although the PESQ scores across the three types of perturbation have been lowered to the average of 1.4, it remains within the perceptually acceptable range in terms of audibility. This shows that simple audio manipulation through editing can be effective in removing the ad hoc watermark, albeit it comes with acceptable but noticeable degradation in audio quality.

B. PGD

Figure 3 shows the relationship between the perceptual quality of the adversarial audio (measured by PESQ) and the

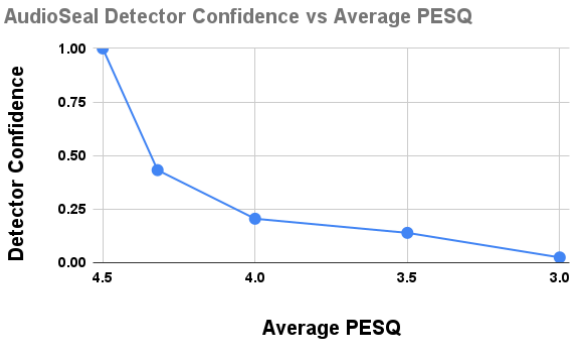

Fig. 3. Trade-off between adversarial attack strength and perceptual audio quality for the PGD attack. As the perturbation strength increases, detector confidence decreases while PESQ also degrades.

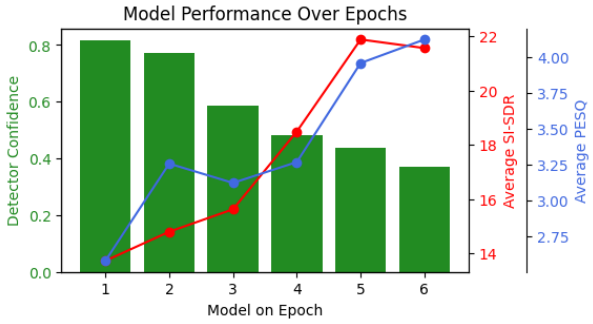

Fig. 4. Model performance across training epochs. Green bars indicate the AudioSeal detector confidence, while the red and blue curves represent SI-SDR and PESQ respectively. During early epochs the model reduces detector confidence primarily by introducing noise-like perturbations. In later epochs the model learns to remove watermark artifacts while restoring audio quality.

resulting detector confidence. As the perturbation strength increases, the detector confidence decreases significantly. However, this improvement in attack effectiveness comes at the cost of reduced perceptual audio quality.

We observe that modest perturbations can already reduce the detector confidence substantially while maintaining acceptable PESQ scores. Stronger attacks further suppress the detector output but introduce increasingly noticeable distortions in the audio signal. Notably, even a small degradation in perceptual quality (e.g., PESQ ≈ 4.3) is sufficient to reduce the detector confidence to below 0.5. Empirically, we found that audio with PESQ scores below 4 becomes noticeably perturbed.

One notable observation is that the watermark appears to introduce audible static-like noise in the signal. Since PESQ is computed relative to the watermarked audio rather than the original clean audio, the metric implicitly encourages preserving this watermark-induced noise. As a result, adversarial attacks that attempt to remove the watermark may also reduce PESQ, even if the resulting audio subjectively sounds cleaner.

This observation suggests that denoising approaches may be effective in suppressing watermark signals. To test this hypothesis, we performed a small-scale denoising experiment on 64 audio samples using a publicly available Python background noise removal library [5]. The resulting audio is perceptually cleaner, as the background noise introduced by the watermark is removed. However, the PESQ score relative to the watermarked audio decreases to around 4, since measuring the distance between the two implicitly encourages preserving the watermark-induced noise. In addition, the detector confidence decreases to an average of 0.4263, indicating that denoising may partially remove watermark artifacts from AudioSeal.

C. Autoencoder Purification

Figure 4 shows the evolution of detector confidence, SI-SDR, and PESQ across training epochs. Due to resource bottleneck, the autoencoder is trained for 6 epochs, with 2000 optimization steps per epoch and a batch size of 64. Model checkpoints are saved after each epoch for evaluation. Due to hardware resource limitation, we only performed small scale experiments to choose the hyperparameters. To analyze model behavior during training, checkpoints are evaluated using randomly sampled sets of 64 audio clips containing both clean and watermarked signals. This evaluation set is consistent across all 6 epochs' evaluation.

We observe two distinct phases during training.

a) Phase 1: Detector Evasion (Epochs 1–3): During early training epochs, the autoencoder tends to introduce noise-like perturbations that reduce the watermark detector confidence. While this behavior successfully lowers the detection score, it temporarily degrades perceptual audio quality.

b) Phase 2: Audio Restoration (Epochs 4–6): In later epochs, the model begins removing watermark-related perturbations while restoring the natural structure of the audio signal. Both SI-SDR and PESQ improve significantly, indicating better reconstruction quality.

By the final epoch, the reconstructed audio becomes effectively *purified* relative to the clean (non-watermarked) reference signal, reducing detector confidence from 1 to 0.4 in average. For a fraction of audios, the autoencoder successfully degrades the confidence of the detector to 0. While additional training may further improve results, the autoencoder approach is considerably more computationally expensive than PGD. In contrast, PGD can quickly drive detector confidence close to 0 in the white-box setting. Despite operating without access to the detector model, the autoencoder still successfully suppresses watermark artifacts while preserving perceptual audio quality.

D. Blackbox Transferability Evaluation

Table I summarizes the transferability results across different attack methods. To begin, our statistical attack shows that simple pitch manipulation can partially degrade the detector. Specifically, slowing the audio to $0.9\times$ reduces the detector confidence to 0.58. This transformation lowers the pitch and alters temporal characteristics of the waveform, suggesting that

Strategy	Average PESQ	Detector Confidence
Raw	4.50	1.00
PGD	2.50	0.98
AE (Epoch 6)	2.61	0.565
Speed 0.9× + Pitch Fix	1.23	0.95
Speed 0.9×	1.10	0.58
Sample Rate Change	1.85	0.98
Bitrate Compression	4.33	0.98

TABLE I
TRANSFERABILITY EVALUATION AGAINST THE ELEVENLABS AI SPEECH DETECTOR.

the detector is sensitive to certain pitch-related features of the speech signal.

Our PGD attack does not transfer effectively to the ElevenLabs detector. Although PGD significantly reduces detector confidence in the white-box AudioSeal setting, its effect on the ElevenLabs detector is negligible, with the confidence remaining close to 0.98. This result is somewhat expected, as PGD optimizes perturbations using gradients from the specific white-box model. The perturbations are constrained within an ϵ-bounded region and are tailored to exploit the decision boundary of AudioSeal's detector. As a result, the generated adversarial examples are highly model-specific and fail to generalize to a different detector architecture such as the ElevenLabs model.

Our autoencoder purification attack, on the other hand, significantly reduces detector confidence to an average of 0.565 while maintaining reasonable perceptual quality (PESQ $= 2.61$). Unlike PGD, which relies on gradients from a specific white-box model, the autoencoder operates as a black-box transformation on the waveform. As a result, it makes fewer assumptions about the underlying detector and instead learns a general purification function that removes watermark-related artifacts from the signal. This broader objective allows the attack to generalize better across different models, leading to improved transferability to the ElevenLabs detector.

Initially, the attack produced weaker results because the autoencoder was trained on audio sampled at 16 kHz, whereas ElevenLabs audio is typically generated at 44.1 kHz. To address this mismatch, we first downsample the audio to 16 kHz for processing by the autoencoder and then resample it back to the original rate. After this adjustment, the purification attack becomes effective. Additionally, this transformation is justifiable since our early experiment shows that changing sample rate does not affect the detector confidence, on both AudioSeal and ElevenLabs.

Finally, the transferability results suggest that the autoencoder is not merely acting as a denoiser specific to AudioSeal. The success of the autoencoder indicates that the learned representation removes broader watermark-related artifacts rather than simply suppressing AudioSeal's watermark, thus not overfitted. This also suggests that the watermarking techniques used by AudioSeal and ElevenLabs may share similar signal characteristics; if the two watermarking schemes were fundamentally different, the purification model would be less likely to transfer successfully. However, the 56% transfer confidence suggests that ElevenLabs may combine multiple techniques to watermark their audio such that if one watermarking scheme failed, there is still another watermark in the audio that is still recognizable. This idea traces back to using ensemble machine learning model. However, more work needs to be done to verify this hypothesis.

V. DISCUSSION

First, adversarial research in audio still seems relatively underdeveloped compared to images. Audio watermarking appears particularly sensitive to simple signal manipulations such as denoising, speed perturbation, and pitch modification. These operations are cheap, widely available, and easy to apply using standard audio processing tools, yet they can substantially reduce detector confidence. This suggests that many current watermarking approaches may be too tightly coupled to low-level signal details that are not stable under realistic editing pipelines.

A related and somewhat alarming observation is that several successful attack vectors do not require sophisticated optimization at all. Simple signal transformations such as denoising or slight pitch and speed adjustments already degrade detector confidence. These operations are inexpensive, widely available, and can be performed using standard audio processing tools (e.g., `ffmpeg` or common denoising libraries). As a result, they represent practical and cost-effective attack vectors against current watermarking systems. Our experiments suggest that these inexpensive transformations may already be sufficient to suppress watermark signals, indicating that access to the detector model itself may not be necessary to mount effective attacks, at least for the two systems we evaluated.

Notably, such attacks do not require specialized expertise. An adversary could perform them using common audio editing software without deep technical or machine learning knowledge. This is particularly concerning in the audio domain, where signals are frequently layered or mixed (e.g., background music, narration, or environmental sound). Overlaying audio may naturally interfere with watermark signals, and understanding the impact of such layering remains an interesting direction for future investigation.

Our experiments also suggest that evaluation methodology matters as much as the attack itself. In particular, if metrics such as PESQ are computed relative to the watermarked signal, they may implicitly reward preserving watermark-induced artifacts. In our case, the watermark was often audible as static-like noise, even though the quality metrics remained relatively high. This means a watermarking system can appear strong quantitatively while still failing an important perceptual test: whether the watermark is actually unobtrusive to human listeners. Therefore incorporating human evaluation into the loop may therefore be important for a more holistic assessment of watermark quality.

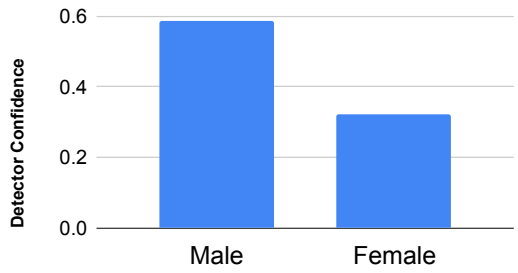

Fig. 5. Detector confidence after purification grouped by speaker gender. Lower values indicate more successful watermark removal. Watermarks in female voices are easier to remove.

More broadly, our findings reflect an inherent asymmetry in security design. It is often significantly easier to break a system than to design one that remains robust under many real-world conditions. In our experiments, disrupting the watermark with relatively small perturbations was straightforward, whereas designing a watermark that is both imperceptible and robust to signal transformations appears substantially more challenging.

Finally, we hypothesize that the vulnerability lies in the fundamental nature of embedded verification watermarks, which can be manipulated with relative ease and little technical knowledge. To counteract, one possible future research direction could be establishing a public verification infrastructure from the watermark service, in which the platform publishes which work has been watermarked and the lack thereof would flag it as manipulated.

As a small side note, the attack appears slightly more successful for female voices (Figure 5). Female voices typically have higher fundamental frequencies and different spectral characteristics compared to male voices. If the watermark interacts with these features, the effectiveness of purification may vary depending on the speaker's vocal characteristics. This suggests that some types of audio may be inherently easier to watermark or make robust than others, analogous to hiding perturbations in textured regions (e.g., painting glaze) where they are less perceptible.

VI. Conclusion

In this paper, we demonstrate the vulnerability of two state-of-the-art audio watermarking schemes through simplistic adversarial attacks, showing that audio watermarking is much more sensitive to signal changes such as noise removal, speed, or pitch adjustments. In addition, white-box attacks are easier to execute but do *not* necessarily transfer to unseen models. In contrast, black-box learned attacks, such as autoencoder purification, generalized better and transferred more effectively across detectors. We stress the point that easier to break a system than to design one that remains robust under many conditions. In our case, disrupting a watermark with small perturbations was relatively straightforward, while designing a well-rounded yet robust watermark is significantly harder.

VII. Contributions

Member	Contribution
Eddie	Statistical Attack + Evaluation
Rani	Autoencoder Attack + Evaluation
William	PGD Attack + Evaluation

TABLE II
Division of work among team members.

References

[1] Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi. High fidelity neural audio compression. *arXiv preprint arXiv:2210.13438*, 2022.

[2] Daniel Galvez, Greg Diamos, Juan Ciro, Juan Felipe Cerón, Keith Achorn, Anjali Gopi, David Kanter, Maximilian Lam, Mark Mazumder, and Vijay Janapa Reddi. The people's speech: A large-scale diverse english speech recognition dataset for commercial usage, 2021. URL: https://arxiv.org/abs/2111.09344, https://arxiv.org/abs/2111.09344 arXiv:2111.09344.

[3] Mohamad Jawhar, Houssein Mourad, Majed Almotairi, and Shadi Jawhar. Addressing the rise of ai-generated misinformation: Challenges and implications. In *2025 IEEE/ACIS 23rd International Conference on Software Engineering Research, Management and Applications (SERA)*, pages 289–294. IEEE, 2025.

[4] A.W. Rix, J.G. Beerends, M.P. Hollier, and A.P. Hekstra. Perceptual evaluation of speech quality (pesq)-a new method for speech quality assessment of telephone networks and codecs. In *2001 IEEE International Conference on Acoustics, Speech, and Signal Processing. Proceedings (Cat. No.01CH37221)*, volume 2, pages 749–752 vol.2, 2001. https://doi.org/10.1109/ICASSP.2001.941023 doi:10.1109/ICASSP.2001.941023.

[5] Tim Sainburg, Marvin Thielk, and Timothy Q Gentner. Finding, visualizing, and quantifying latent structure across diverse animal vocal repertoires. *PLoS computational biology*, 16(10):e1008228, 2020.

[6] Robin San Roman, Pierre Fernandez, Hady Elsahar, Alexandre D´efossez, Teddy Furon, and Tuan Tran. Proactive detection of voice cloning with localized watermarking. *ICML*, 2024.